Chapter 31
Short-Rate Models

Chapter 29 priced a cap and a swaption with Black's model and Chapter 30 corrected the forward when the payoff arrives at the wrong time or in the wrong currency. Both took a quoted volatility and produced one number.

Neither built a model of the term structure, and there is a large class of instruments for which that is fatal. A **Bermudan swaption** can be exercised on any one of a fixed set of dates; its holder decides at year three whether to exercise, and that decision depends on the entire curve at year three—a curve Black's model never constructs. A callable bond's issuer faces the same decision annually for a decade. A structured note pays a coupon that depends on where rates are, conditional on where they have been.

For these, the model has to produce a consistent evolution of the whole curve, not a distribution for one rate at one date. A desk running a book needs more than that: one model that prices every deal on it consistently, so that a swaption, a callable bond and a Bermudan are valued off the same curve dynamics and their risks aggregate. The oldest and still most widely used way to do that is to model the short rate and derive everything else from it.

Section 31.12 is the one that changes how the rest of Part VI reads: it prices the defect that motivates Chapter 32, and it does so with a number rather than an assertion. A one-factor model prices a CMS spread option at 56.2 basis points when, at a realistic correlation between the two rates, it is worth 245.0.

Sections 31.1 to 31.4 are the equilibrium models, Vasicek and Cox–Ingersoll–Ross (CIR), and why a model that cannot reprice today's curve is not used to price. Sections 31.5 to 31.8 are the no-arbitrage answer: Ho–Lee, Hull–White, and the tree that implements it. Sections 31.9 to 31.11 are what the tree is built for: Bermudan swaptions, what calibration cannot pin down, and callable bonds through option-adjusted spread. Sections 31.12 to 31.15 are the repairs and their limits: the one-factor failure priced in basis points, two factors, negative rates, and the transition; Sections 31.16 to 31.18 say where these models are still the right tool and who uses them.

31.1 WHAT A SHORT-RATE MODEL IS

Specify a stochastic process for the instantaneous short rate $r(t)$ (the rate for borrowing over the next instant, the limit of the zero rate as maturity shrinks to zero) under the risk-neutral measure. Every zero-coupon bond price then follows from

$$P(t,T) = E^{Q}\left[\exp\left(-\int_{t}^{T} r(s)\,ds\right)\,\middle|\,\mathcal{F}_t\right] \tag{31.1}$$

which is Chapter 13's risk-neutral valuation applied to a payoff of \$1 at time T.

This is a strong modeling choice. The entire term structure, at every future date, is generated by a single scalar state variable. That is the source of the approach's tractability and of the defect in Section 31.12, and the two cannot be separated. Two properties make a model usable.

Analytic bond prices. If (31.1) evaluates in closed form, the model is fast enough to calibrate. The most widely used class for which it does is the **affine** models, in which $\ln P(t,T)$

is affine in r:

$$P(t,T) = A(t,T)\, e^{-B(t,T)\, r(t)} \tag{31.2}$$

Written in rates rather than prices, with $R(t,T)$ the continuously compounded zero rate of Section 4.8,

$$R(t,T) = -\frac{\ln P(t,T)}{T-t} = \frac{B(t,T)\, r(t) - \ln A(t,T)}{T-t}$$

so every rate on the curve is affine in the one state variable, and Example 31.1's curves follow from it.

And the ability to fit today's curve. A model whose output curve differs from the observed one misprices the underlying swaps and bonds themselves, before any option is considered. This requirement, not realism, is what separated the two generations of models, and Section 31.4 makes the separation concrete.

31.2 VASICEK AND THE EQUILIBRIUM APPROACH

The first generation asked what process for r is economically reasonable, and derived the curve from it. Vasicek's model is

$$dr = a(b - r)\, dt + \sigma\, dW \tag{31.3}$$

The drift pulls r toward b at speed a. This is **mean reversion**, the property that distinguishes rates from equities. A stock price has no level it returns to; a short rate does, because a central bank and an economy both act on it. Figure 31.1 shows what reversion does to the whole distribution, not just to the expected rate. Reversion is slow at plausible speeds: the half-life, the time over which the expected gap between r and b halves, is $\ln 2/a$, about seven years at $a = 0.10$, so over the life of a ten-year deal the pull toward b is partial rather than complete. Set $a = 0$ and the rate is a driftless random walk whose standard deviation grows as $\sigma\sqrt{t}$ without limit; restore a and it converges to $\sigma/\sqrt{2a}$. That limit is what lets the model produce a finite long rate. Note which rate reverts to b: the expected short rate does, as t grows, while the zero curve's long end sits at the lower level R_∞ marked on the figure, whatever the short rate happens to be.

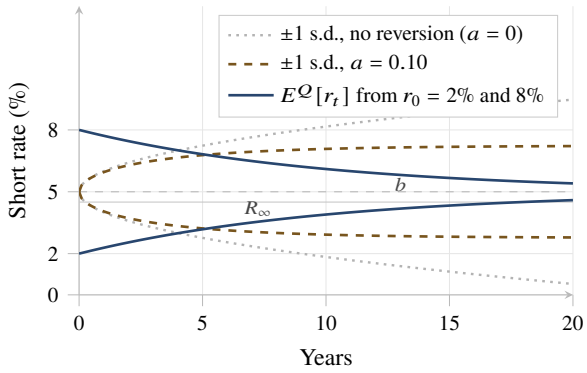


Figure 31.1: Mean reversion at $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, against the same model with $a = 0$; the bands are ±1 standard deviation from $r_0 = b$, and R_∞ is the limiting zero rate of Example 31.1.

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Equation (31.3) is affine, with

$$B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

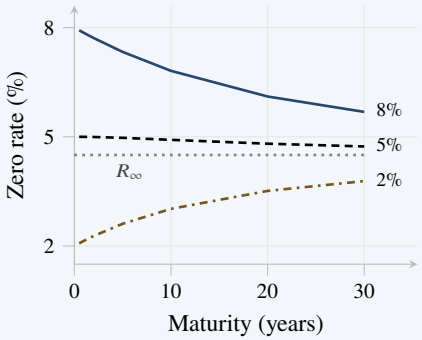
$$\ln A(t,T) = \frac{(B(t,T) - (T-t))\left(a^2 b - \sigma^2/2\right)}{a^2} - \frac{\sigma^2 B(t,T)^2}{4a}$$

Bond prices, and therefore the whole curve, follow from $r(t)$ and three constants. The two terms in σ^2 are the convexity correction. The $-\sigma^2/2$ inside the first is what pulls the long end of the curve below b in Example 31.1; the last term is bounded, since $B(t,T)$ tends to $1/a$, so once divided by maturity its contribution to the zero rate dies away.

Example 31.1 Three starting rates, one curve

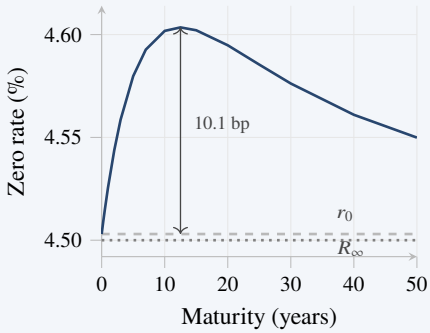
Take $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, and compute the zero curve from three starting short rates.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.073%	5.000%	7.926%
1y	2.144%	4.998%	7.853%
2y	2.275%	4.994%	7.713%
5y	2.610%	4.971%	7.332%
10y	3.020%	4.916%	6.812%
20y	3.513%	4.810%	6.107%
30y	3.783%	4.734%	5.684%



The same parameters give an upward-sloping curve and a downward-sloping one. Even at $r_0 = b$ the curve falls, by 27 basis points over thirty years, because convexity pulls it toward R_∞ rather than b.

A third shape is available, but barely. The curve is humped for $b - 3\sigma^2/(4a^2) < r_0 < b$, a band 75 basis points wide at these parameters, and the largest hump anywhere in that band is about 10 basis points, at $r_0 = 4.50\%$ peaking at twelve and a half years. Real curves hump by five to ten times that. The shape is in the model; the magnitude is not.



$a = 0.10$, $b = 5\%$, $\sigma = 1\%$, $r_0 = 4.503\%$

All three converge toward the same long rate, and it is not b:

$$R_\infty = b - \frac{\sigma^2}{2a^2} = 5\% - \frac{0.01^2}{2 \times 0.10^2} = \mathbf{4.50}\%$$

The gap is a convexity effect, and it is the same effect as Chapter 30's: a long bond's price is convex in the rate, so uncertainty raises its expected price and lowers its yield. Here the model produces the adjustment automatically instead of requiring it to be added.

Three parameters and one state variable generate the entire curve. The curve has hundreds of observable points and the model has four numbers, so it cannot match the market. That is not a calibration failure; it is arithmetic.

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31.3 COX–INGERSOLL–ROSS

Vasicek's r is normally distributed and can go negative. The best-known repair is

$$dr = a(b - r)\,dt + \sigma\sqrt{r}\,dW \tag{31.4}$$

The $\sqrt{r}$ has two consequences. Volatility falls as the rate falls, as US short-rate data before 2008 showed, though Chan, Karolyi, Longstaff and Sanders (1992) estimate a stronger dependence on the level than the square root; and the diffusion coefficient vanishes at zero, so the rate remains nonnegative. Strict positivity requires more, namely the **Feller condition** $2ab > \sigma^2$; when it fails, zero is attained and acts as a reflecting boundary.

The model remains affine, so bond prices retain a closed form, and its coefficients are more involved than Vasicek's without being harder to evaluate. With $\tau = T - t$ and $\gamma = \sqrt{a^2 + 2\sigma^2}$,

$$B(t,T) = \frac{2\,(e^{\gamma\tau} - 1)}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}$$

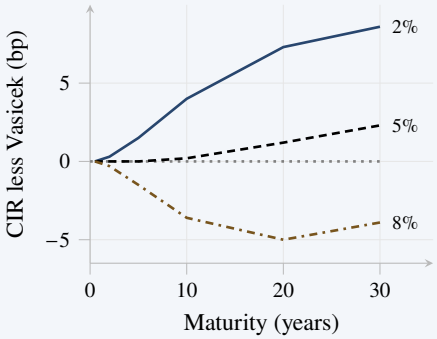
$$A(t,T) = \left[\frac{2\gamma\, e^{(a+\gamma)\tau/2}}{(\gamma + a)\,(e^{\gamma\tau} - 1) + 2\gamma}\right]^{2ab/\sigma^2}$$

The limiting zero rate is $R_\infty = 2ab/(\gamma + a)$, again below b and again by a convexity term, but one that now scales with b, because the volatility scales with the rate. Example 31.2 measures what the change of diffusion does to the curves.

Example 31.2 What positivity costs

The two models are comparable only if their volatilities are. Vasicek's diffusion is σ and CIR's is $\sigma\sqrt{r}$, so the two agree at $r = b$ when CIR's σ is set to $0.01/\sqrt{0.05} = 4.47\%$. Take Example 31.1's $a = 0.10$ and $b = 5\%$ with that value, so both models carry a 1% instantaneous volatility at $r = 5\%$. The Feller condition holds comfortably, $2ab = 0.0100$ against $\sigma^2 = 0.0020$.

Maturity	$r_0 = 2\%$	$r_0 = 5\%$	$r_0 = 8\%$
0.5y	2.074%	5.000%	7.926%
1y	2.144%	4.998%	7.852%
2y	2.278%	4.994%	7.710%
5y	2.626%	4.971%	7.317%
10y	3.059%	4.918%	6.777%
20y	3.586%	4.821%	6.057%
30y	3.869%	4.757%	5.645%



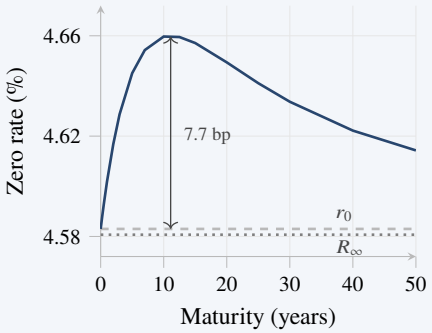
Left: CIR's zero curves. Right: those curves less Example 31.1's, in basis points.

The two sets of curves are nearly the same. The largest gap across thirty years is 8.6 basis points, and inside five years it is under two. Positivity cost almost nothing in the shape of the curve, which is why CIR held the field for two decades.

The gap that does appear has an instructive sign. From $r_0 = 2\%$ the CIR curve sits above Vasicek's, and from 8% below. A convexity correction grows with variance, and in CIR the variance moves with the rate: from a low start there is less of it to depress the long end, from a high start more, while Vasicek applies the same correction either way. The limiting zero rate records the same thing, $R_\infty = 4.58\%$ against Vasicek's 4.50%, a drag below b of 0.42 percentage points rather than 0.50.

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CIR humps too, and over a narrower band: from 4.40% to b, 60 basis points against Vasicek's 75. The largest hump in it is about 8 basis points, at $r_0 = 4.58\%$ peaking at eleven years. Both models put their best hump at R_∞ and neither can make it large. Producing a realistic hump is not something a one-factor model does, whichever diffusion it carries.



$a = 0.10$, $b = 5\%$, $\sigma = 4.47\%$, $r_0 = 4.583\%$

The CIR model was for two decades the standard answer to Vasicek's negative rates, and what happened to that argument should be stated plainly. Rates went negative, in several major currencies, for years. A model that cannot produce a negative rate cannot be calibrated to a market that has one, even approximately, because there is no parameter value under which a process that cannot fall below zero assigns any probability to the observed state.

So the property that made CIR attractive is the property that retired it as a model of rates in those currencies. The process itself remains in wide use wherever the variable cannot be negative: as the default intensity in credit models, and as the variance in Heston's model of Section 21.4. Section 31.14 takes the point up in general, because it recurs.

31.4 WHY EQUILIBRIUM MODELS ARE NOT USED TO PRICE

Example 31.1 contains the whole argument, and it deserves to be stated as a principle rather than a limitation of one model. An **equilibrium model** treats today's term structure as an output. It asks what curve the assumed process implies, and the answer is a curve with three or four parameters' worth of freedom.

A trading desk needs today's term structure to be an input. If the model says the ten-year zero is 4.92% and the market says 4.10%, then every instrument priced on the model is wrong by an amount that has nothing to do with the option in it. A swaption priced on a curve that is eighty basis points off is not a swaption price.

The distinction is worth being precise about, because it is often stated as though equilibrium models were merely less accurate. They are answering a different question, and it is a legitimate one. An economist asking what the curve *should* look like, or a risk manager generating scenarios that are internally consistent over decades, wants a model whose curve is derived rather than fitted. Real-world scenario generation for pension and insurance liabilities still uses models of this kind, for exactly that reason.

But a desk quoting a price cannot use one, and the second generation of models exists to remove this problem and nothing else.

31.5 HO–LEE AND THE NO-ARBITRAGE IDEA

Make the drift time-dependent and choose it so the model reproduces the observed curve. A model built this way, taking today's curve as an input rather than producing it, is a **no-arbitrage model**.

Start from what is observable. The input is today's zero curve, the discount factors $P(0,t)$ that Chapter 7 builds from quoted instruments. What a fitted model needs from them is not the zero rate $R(0,t)$ of Section 31.1 but the **initial forward curve**, the instantaneous rate for

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borrowing at t that today's prices imply:

$$f(0,t) = -\frac{\partial \ln P(0,t)}{\partial t} \qquad \text{equivalently} \qquad P(0,t) = \exp\left(-\int_0^t f(0,u)\,du\right)$$

The two descriptions carry the same information, and $R(0,t)$ is the average of $f(0,u)$ over $[0,t]$. The forward curve is the useful one here because a drift is a local quantity, so it has to be matched against a local description of the curve rather than against an average.

The **Ho–Lee model** is the simplest case:

$$dr = \theta(t)\,dt + \sigma\,dW \tag{31.5}$$

The drift $\theta(t)$ is not a parameter to estimate; it is solved for:

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t$$

and the model fits the curve exactly by construction. This is the whole idea of the no-arbitrage generation, and the surprising part is how cheap it is. One function of time, chosen mechanically, converts a model that cannot price into one that can. The cost is that $\theta(t)$ carries no information (it is whatever it has to be), so a model in this class explains nothing about why the curve has the shape it has. It is an interpolation device with a stochastic component, and that is the correct way to understand every model in the rest of this chapter.

Ho–Lee has one defect that rules it out in practice: *no mean reversion.* The volatility of the T-maturity forward rate is σ regardless of T, so the model says a thirty-year forward rate is as volatile as a one-year forward rate. Section 31.6 shows what mean reversion does to it.

31.6 HULL–WHITE

Add mean reversion to Ho–Lee, or a fitted drift to Vasicek (the same model either way):

$$\boxed{dr = \big[\theta(t) - ar\big]dt + \sigma\,dW} \tag{31.6}$$

As in Ho–Lee, θ is solved for rather than estimated, and in terms of the same initial forward curve $f(0,t)$,

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + a\,f(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right)$$

which returns Section 31.5's θ as $a \to 0$, the last term tending to $\sigma^2 t$. [TN31] derives this and explains why a single function of time can match every maturity at once. In practice it is never evaluated, because bond prices can be written with today's curve in them directly:

$$\ln A(t,T) = \ln \frac{P(0,T)}{P(0,t)} + B(t,T)\,f(0,t) - \frac{\sigma^2}{4a}\left(1 - e^{-2at}\right)B(t,T)^2$$

so θ disappears from everything that is computed. It was solved for the curve, and the curve is what remains.

This, the **Hull–White model**, is the workhorse. It fits the initial curve exactly, has closed-form bond and European option prices, and builds on a recombining tree. Almost every desk that uses a short-rate model uses this one or a close relative. The two constants do different jobs, and confusing them is a common error in using the model.

Example 31.3 What a and σ separately control

The instantaneous forward rate follows by differentiating (31.2):

$$f(t,T) = -\frac{\partial \ln P(t,T)}{\partial T} = -\frac{\partial \ln A(t,T)}{\partial T} + e^{-a(T-t)}\,r(t)$$

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since $\partial B/\partial T = e^{-a(T-t)}$. Only r is stochastic, so the forward rate inherits its diffusion scaled by that factor, and its volatility seen from today is

$$\sigma\, e^{-aT} \tag{31.7}$$

T	$a = 0.01$	$a = 0.05$	$a = 0.10$	$a = 0.20$
1y	99.0	95.1	90.5	81.9
2y	98.0	90.5	81.9	67.0
5y	95.1	77.9	60.7	36.8
10y	90.5	60.7	36.8	13.5
20y	81.9	36.8	13.5	1.8
30y	74.1	22.3	5.0	0.2

Forward-rate volatility in basis points, at $\sigma = 1\%$.

σ sets the level of the volatility term structure and a sets its shape. At $a = 0.01$ the curve declines slowly, close to Ho–Lee: a thirty-year forward is still three-quarters as volatile as a one-year forward. At $a = 0.20$ the thirty-year forward has essentially no volatility at all, which is equally wrong in the other direction.

Because the two act on different features, they are separately identified by the data. Section 29.9's volatility cube contains caps and swaptions of many expirations, and the term structure of their volatilities determines a while their level determines σ.

That is the encouraging version. Section 31.10 shows that for the instruments that matter most, it is not true.

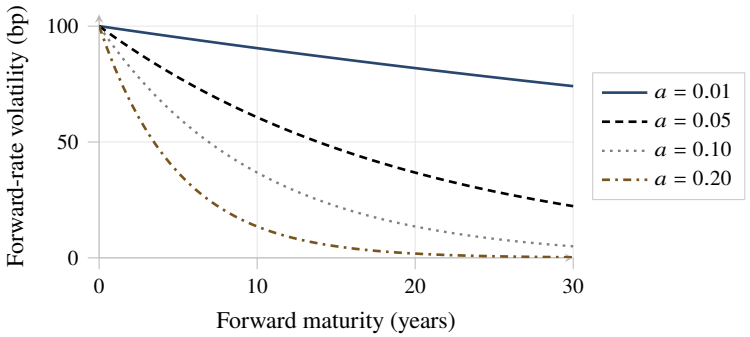


Figure 31.2: Equation (31.7) at $\sigma = 1\%$.

Figure 31.2 draws the table: a alone decides how fast forward-rate volatility dies away with maturity.

31.7 EUROPEAN OPTIONS IN CLOSED FORM

The model is usable before any lattice is built: every European option on bonds, caps and swaptions has a closed form, and all of them come from the zero-bond option formula. Example 31.4 carries these formulas through a cap and a swaption.

An option on a zero-coupon bond. Under the T-forward measure the forward bond price $P(t,S)/P(t,T)$ is a martingale, and since (31.2) makes $\ln P$ affine in a Gaussian r, it is a lognormal one. Its instantaneous volatility $\sigma e^{-a(T-t)}(1 - e^{-a(S-T)})/a$ is deterministic, so the variance accumulates to a number rather than a random variable, and its square root over the option's life is

$$\sigma_P(T,S) = \frac{\sigma}{a}\left(1 - e^{-a(S-T)}\right)\sqrt{\frac{1 - e^{-2aT}}{2a}} \tag{31.8}$$

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Lognormality with a deterministic volatility is the assumption on which Black's formula rests, so for a strike X and an expiration T on the bond maturing at S, the zero-bond call (ZBC) and put (ZBP) are

$$\text{ZBC} = P(0,S)\,N(h) - X\,P(0,T)\,N(h-\sigma_P) \tag{31.9}$$

$$\text{ZBP} = X\,P(0,T)\,N(\sigma_P - h) - P(0,S)\,N(-h) \tag{31.10}$$

$$h = \frac{1}{\sigma_P}\ln\frac{P(0,S)}{X\,P(0,T)} + \frac{\sigma_P}{2}$$

the put (31.10) following from the call (31.9) by parity. Only today's curve and the two parameters appear. Both factors of (31.8) saturate (at σ/a in the bond's remaining life, at $1/\sqrt{2a}$ in the option's), which is (31.7) seen from the other side. [TN31] derives (31.8) and works out both ceilings.

A cap. Section 29.3 showed that a caplet on notional L, struck at K on the rate for the accrual period $[T_i, T_{i+1}]$ of length τ_i, is $L(1+\tau_i K)$ puts on the T_{i+1} bond struck at $1/(1+\tau_i K)$. Substituting (31.10) prices a cap one period at a time, with no approximation anywhere.

A swap. It needs no formula, being linear in bond prices; for the payer of fixed,

$$V = L\left[P(0,T_0) - P(0,T_n) - K\sum_{i=1}^{n}\tau_i P(0,T_i)\right]$$

and neither a nor σ appears. A swap's value comes entirely from the curve, which the model reproduces by construction. A model's parameters do work only where there is optionality, and a model that misprices a swap has made a fitting error rather than a modeling one. That is Section 31.4's argument in one line.

A European swaption. Exercising a payer swaption is exercising the right to sell, at par, a bond paying $c_i = K\tau_i$ at each t_i, with the principal added to the last payment, $c_n = 1 + K\tau_n$; that is the equivalence of Section 29.6. What makes it tractable here is the single factor. Each $P(T,t_i)$ is a decreasing function of $r(T)$, so the coupon bond is too, and there is a unique r^* at which it is worth par. Above r^* *every* component bond is simultaneously below its value $X_i = P(T,t_i;r^*)$ there, and below r^* every one is above it. The exercise regions coincide, so

$$\text{payer swaption} = L\sum_{i=1}^{n} c_i\,\text{ZBP}(T,t_i,X_i) \tag{31.11}$$

Equation (31.11) is **Jamshidian's decomposition**, and it is an identity rather than an approximation. It rests on monotonicity in a single state variable, so it survives neither a second factor (where the exercise region is a curve in the plane rather than a threshold) nor a Bermudan schedule, where the decision at one date turns on the value of the rights that remain. Both go to the tree of Section 31.8.

Example 31.4 A cap and a swaption

Take a flat continuously compounded curve at 4%, so $P(0,T) = e^{-0.04T}$, with $a = 0.08$ and $\sigma = 1\%$ on a notional of 100. Left: a two-year semiannual cap struck at 4%, every period's forward rate being 4.040268%. Right: a 1×3 annual payer swaption struck at the par rate, whose annuity is 2.662181 and par rate 4.081077%, and $r^* = 3.987895\%$, the short rate at which the coupon bond is worth par.

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Caplet	σ_P	Value
[0.5, 1.0]	33.98	0.142743
[1.0, 1.5]	47.12	0.190218
[1.5, 2.0]	56.60	0.222055
Cap		**0.555017**

Maturity	c_i	X_i	Value
2y	0.040811	0.960860	0.014025
3y	0.040811	0.923177	0.025775
4y	1.040811	0.886915	0.906913
Swaption			**0.946713**

σ_P in basis points.

The caplets rise in value with their expiration although the forward rate is identical in each, because σ_P grows with the option's life, and that growth is bounded, which is mean reversion at work. In the swaption almost all the value sits in the final strip, which carries the principal.

Both figures were also computed a second way, by integrating the payoff directly over the distribution of $r(T)$ with no decomposition and no bond-option formula, and the two routes agree to nine decimal places. That check is only correct under the T-forward measure, where

$$r(T) \sim \mathcal{N}\left(f(0,T),\ \frac{\sigma^2}{2a}\left(1-e^{-2aT}\right)\right)$$

The mean is the forward rate exactly, while under the risk-neutral measure it carries a further $(\sigma^2/(2a^2))(1-e^{-aT})^2$. Integrating the swaption against that instead returns 0.952366 rather than 0.946713, an error of 0.6% from nothing but the wrong measure in the last step, which is the subject of the two examples that follow.

Chapter 28 proved that the choice of numeraire cannot change a price, and that the right choice can change the work enormously. With a model in hand, both claims can be tested.

Example 31.5 Invariance, and what it buys

A caplet on the simple rate over $[2.0, 2.5]$, paid at 2.5, struck at 4%, notional 100. The economy is (31.6) with $a = 0.08$ and $\sigma = 1\%$, on the chapter's market curve, $R(0,T) = 4.2\% - 1.2\%\, e^{-0.25T}$ continuously compounded, so that $P(0, 2.0) = 0.932913$ and $P(0, 2.5) = 0.914898$. The forward rate follows:

$$F_0 = \frac{1}{\tau}\left(\frac{P(0,T_1)}{P(0,T_2)} - 1\right) = 2.0 \times \left(\frac{0.932913}{0.914898} - 1\right) = 3.9381\%$$

(a) Under the 2.5-year forward measure. The forward rate is a martingale (no drift to model), and by the identity of Section 29.3, with $1 + \tau K = 1 + 0.5 \times 4\% = 1.02$, the caplet is $L(1 + \tau K) = 102$ puts on the 2.5-year bond struck at $1/(1 + \tau K) = 0.980392$. That bond is lognormal here, because (31.2) makes $\ln P$ affine in r and Hull–White's r is Gaussian; its forward price carries the deterministic volatility (31.8):

$$v = \sigma_P(T_1,T_2) = \frac{0.01}{0.08}\left(1-e^{-0.08\times 0.5}\right)\sqrt{\frac{1-e^{-2\times 0.08\times 2}}{2\times 0.08}} = 64.12 \text{ bp}$$

As with the zero-bond options of (31.10), that makes Black's formula exact here rather than an approximation:

$$\text{caplet} = L \cdot P(0,T_2)\,[(1+\tau F_0)N(d_1) - (1+\tau K)N(d_2)] = \mathbf{0.22478353}$$

with $d_{1,2} = \ln[(1+\tau F_0)/(1+\tau K)]/v \pm v/2$: a closed form, evaluated in microseconds.

(b) Under the risk-neutral measure. Now the numeraire is the money market account, so the payoff must be discounted along each simulated path of r. Two million antithetic paths of two hundred steps each give

$$\mathbf{0.22498411} \qquad \text{(standard error } 0.00023576\text{)}$$

The two differ by 0.85 standard errors. This is the theorem, confirmed: the numeraire is a choice of units and prices do not depend on it.

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(c) The error the measure change exists to prevent. Suppose one computes the expected payoff under the risk-neutral measure and discounts it at today's zero rate, treating the payoff and the discount factor as independent:

$$0.23044665 \qquad \text{an error of } +\mathbf{2.52}\%$$

That 2.52% is the covariance between the payoff and the discount factor, as a fraction of the price. A caplet pays when rates are high, and when rates are high the discount factor is small. Ignoring that correlation overstates the value.

The forward measure removes this error: using $P(t,T_2)$ as numeraire folds the discounting into the numeraire, so no covariance term survives. Black's formula for a caplet is exact rather than approximate because the convention already made this choice.

The same economy settles a second question. A forward price is an expectation under the T-forward measure, as Section 28.7 shows, and is not one under the risk-neutral measure; the difference is a covariance.

Example 31.6 A forward price is not a risk-neutral expectation

A non-dividend-paying stock, $S_0 = 100$, volatility 25%, $T = 5$, with the curve of Example 31.5 so that $P(0,T) = 0.824639$. The forward price is

$$F(0,T) = \frac{100}{0.824639} = 121.2652$$

and this is a no-arbitrage fact: it does not depend on any correlation.

Now compute $E^{Q}[S_T]$ in the Hull–White economy of Example 31.5 for a range of correlations between the stock and the short rate. That expectation has a name. A futures price is a martingale under Q, as Section 28.7 established, so $E^{Q}[S_T]$ *is* the futures price for delivery at T, and the ratio below is the futures–forward basis of Section 30.3, computed inside a model rather than read off a screen:

$\rho(S,r)$	$E^{Q}[S_T]/F(0,T)$	**Gap**
−0.50	0.989442	−1.056%
−0.25	0.996260	−0.374%
0.00	1.003125	+0.313%
+0.25	1.010037	+1.004%
+0.50	1.016997	+1.700%

The forward price is fixed and $E^{Q}[S_T]$ moves with ρ, so with stochastic rates the two agree only at one particular correlation, by coincidence; in general they coincide only when rates are deterministic. Zero correlation is not enough: the $\rho = 0$ row still runs 0.31% high.

The gap has two parts, the two terms of the exponent in

$$\frac{E^{Q}[S_T]}{F(0,T)} = \exp\left[\mathrm{Var}^{Q}\left(\int_0^T r\,ds\right) + \frac{\rho\,\sigma_S\,\sigma}{a}\big(T - B(0,T)\big)\right]$$

with σ_S the equity volatility; [TN31] derives it. The first term survives at $\rho = 0$ and is the discount factor's own convexity: the stock grows at the short rate, and $E^{Q}[e^{\int_0^T r\,ds}] = e^{\mathrm{Var}^{Q}(\int_0^T r\,ds)}/P(0,T)$, which exceeds $1/P(0,T)$ whenever the short rate is stochastic. The second moves with ρ, through the stock's own shocks. Together they are the covariance between the asset and the discount factor, the same kind of quantity that produced the 2.52% error in Example 31.5: a caplet mispriced by ignoring that covariance and a futures price standing apart from a forward are one effect under two names.

A practical warning follows. A Monte Carlo engine that simulates under the risk-neutral measure and reports "the expected terminal price" is not reporting the forward; it is reporting the futures price. At $\rho = 0.5$ over five years the difference is 1.7%, which is larger than the bid–ask on the forward it is being compared with.

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31.8 BUILDING THE TREE

The closed forms of Section 31.7 cover the European cases. Everything else needs a lattice, and Hull–White's is built in two stages, a split that is the construction's central idea rather than a computational detail.

Stage one: the geometry and the probabilities. Stage one builds a tree for the process with $\theta(t)$ removed, which reverts to zero,

$$dx = -ax\,dt + \sigma\,dW, \qquad x(0) = 0 \tag{31.12}$$

Nodes sit at $x = j\Delta x$. Over one step from node j the move has mean and variance

$$M = -a\,j\Delta x\,\Delta t, \qquad V = \sigma^2\Delta t$$

and the three branching probabilities are fixed by reproducing both, together with the requirement that they sum to one. Writing $\eta = M/\Delta x = -aj\Delta t$ and branching to $j+1$, j, $j-1$,

$$p_u + p_m + p_d = 1, \qquad p_u - p_d = \eta, \qquad p_u + p_d = \frac{V}{\Delta x^2} + \eta^2$$

The spacing is chosen to make this tidy. Setting $\Delta x = \sigma\sqrt{3\Delta t}$ turns $V/\Delta x^2$ into $1/3$, and the three equations solve to

$$p_u = \frac{1}{6} + \frac{\eta^2 + \eta}{2}, \qquad p_m = \frac{2}{3} - \eta^2, \qquad p_d = \frac{1}{6} + \frac{\eta^2 - \eta}{2} \tag{31.13}$$

At the center $\eta = 0$ and these are $1/6$, $2/3$, $1/6$. Nothing about the curve has entered yet, which is why stage one can be built once and reused.

Where symmetric branching breaks down. Write $u = aj\Delta t$, so $\eta = -u$. Both p_u and p_d in (31.13) stay positive for every u (each is a quadratic in u with negative discriminant), but the middle one does not:

$$p_m = \frac{2}{3} - u^2 < 0 \qquad \text{once} \qquad u > \sqrt{2/3} = 0.8165$$

In the node index that reads $j > 0.8165/(a\Delta t)$: far enough above the center, the drift is too strong for symmetric branching to match.

The remedy is to branch downward from the top node, to j, $j-1$, $j-2$. The same three equations then give

$$p_u = \frac{7}{6} + \frac{u^2 - 3u}{2}, \qquad p_m = -\frac{1}{3} - u^2 + 2u, \qquad p_d = \frac{1}{6} + \frac{u^2 - u}{2} \tag{31.14}$$

and again the middle probability is the binding one. Solving $-\frac{1}{3} - u^2 + 2u \geq 0$ gives

$$1 - \sqrt{2/3} \;\leq\; u \;\leq\; 1 + \sqrt{2/3}, \qquad \text{that is} \qquad 0.1835 \leq u \leq 1.8165$$

So each pattern is valid on an interval, and the intervals overlap on $0.1835 \leq u \leq 0.8165$. The switch has to happen somewhere inside that window, and taking it as early as possible keeps the tree narrowest. So $j_{\max}$ is the smallest integer satisfying

$$j_{\max} \;\geq\; \frac{1 - \sqrt{2/3}}{a\,\Delta t} \;=\; \frac{0.1835}{a\,\Delta t} \tag{31.15}$$

that is, the first node at which the switched branching is legal. The bottom of the tree mirrors the top: at $-j_{\max}$ the branching runs flat, one up and two up, with (31.14) read at $-u$ and p_u and p_d exchanged.

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Stage two: displacing the tree onto the curve. Stage two shifts every level i by an $\alpha(i)$, chosen so that $r = x + \alpha$, with x the process of (31.12), reprices today's zero curve. The curve is Example 31.5's; every tree example that follows uses it. The displacements are computed forward, one time step at a time, from the Arrow–Debreu prices. Write $Q(i, j)$ for the **Arrow–Debreu price**, today, of a security that pays \$1 if the tree reaches node (i, j) and nothing otherwise (Section 13.3's state price, under the name the rates literature uses). The displacements are then *solved*, not fitted, and Example 31.7 carries that out in full.

One feature of the geometry is worth a figure, because it is where the construction stops being obvious.

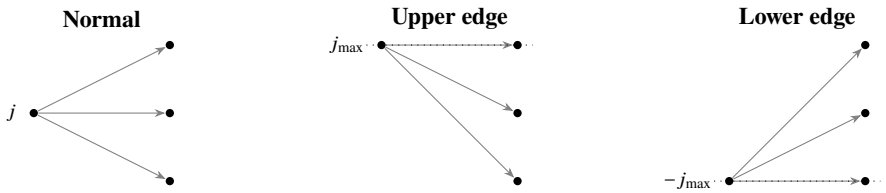


Figure 31.3: Branching switches at $\pm j_{\max}$. Schematic; spacing exaggerated.

Figure 31.3 is (31.15) drawn: below $j_{\max}$ the branching is symmetric, at the top node it runs flat, one down and two down, and at the bottom node flat, one up and two up. The tree therefore stops widening. That is not an approximation—it is what mean reversion means, expressed in the geometry.

Example 31.7 Building the tree, step by step

Take this chapter's curve with $a = 0.08$ and $\sigma = 1\%$, but $\Delta t = 1$ year, so that every number fits on the page. The tree that prices Section 31.9's examples runs the identical construction at 24 steps a year.

Stage one fixes the geometry and never changes. $\Delta x = 0.01\sqrt{3 \times 1} = 0.017321$ and $j_{\max}$ is the smallest integer at or above $0.1835/(0.08 \times 1) = 2.294$, so $j_{\max} = 3$. The branching probabilities depend on j alone (not on the curve, not on the step), because stage one knows nothing about either:

j	p_u	p_m	p_d	**branches to**
+3	0.835467	0.089067	0.075467	$j, j-1, j-2$
+2	0.099467	0.641067	0.259467	$j+1, j, j-1$
+1	0.129867	0.660267	0.209867	$j+1, j, j-1$
0	0.166667	0.666667	0.166667	$j+1, j, j-1$
−1	0.209867	0.660267	0.129867	$j+1, j, j-1$
−2	0.259467	0.641067	0.099467	$j+1, j, j-1$
−3	0.075467	0.089067	0.835467	$j+2, j+1, j$

The drift shows in the asymmetry: above the center the down probability exceeds the up, and the imbalance grows with $|j|$. That is mean reversion, discretized. The two end rows come from (31.14), and their probabilities attach to different nodes, which is why p_u is the largest of the three at $j = +3$, where it is the probability of *staying* at j rather than of rising above it. Four steps is enough to reach them: the tree touches $j = \pm 3$ at level 3, and the switched rows are what carry it from there to level 4.

Stage two solves one equation per step. Begin with $Q(0, 0) = 1$. At level i the displacement that makes the tree reproduce the next discount factor is

$$\alpha(i) = \frac{1}{\Delta t}\left[\ln \sum_j Q(i, j)\, e^{-j\Delta x\, \Delta t} \;-\; \ln P\big(0, (i+1)\Delta t\big)\right] \tag{31.16}$$

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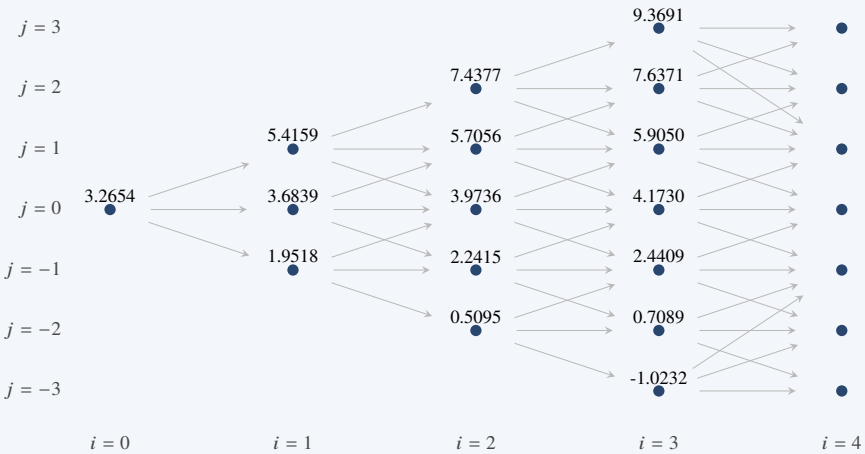
after which the Arrow–Debreu prices roll forward,

$$Q(i+1,k) = \sum_j Q(i,j)\; p(j \to k)\; e^{-(\alpha(i)+j\Delta x)\Delta t}$$

in which $p(j \to k)$ is whichever of p_u, p_m, p_d carries node j to node k, and zero if none of them does, so the *branches to* column above is what decides which. Four steps of that give

	$i=0$	$i=1$	$i=2$	$i=3$	$i=4$
$Q(i,+3)$	—	—	—	0.0018	0.0048
$Q(i,+2)$	—	—	0.0198	0.0369	0.0477
$Q(i,+1)$	—	0.1613	0.2045	0.2092	0.2017
$Q(i,\ 0)$	1.0000	0.6452	0.4799	0.3907	0.3356
$Q(i,-1)$	—	0.1613	0.2081	0.2165	0.2123
$Q(i,-2)$	—	—	0.0205	0.0395	0.0528
$Q(i,-3)$	—	—	—	0.0020	0.0056
$\alpha(i)$	3.2654%	3.6839%	3.9736%	4.1730%	—
$P(0,(i+1)\Delta t)$	0.9679	0.9329	0.8967	0.8604	—

The last column carries no α: level 4 is where this tree stops, and $\alpha(4)$ would be fixed by $P(0,5)$.



Short rates $r(i,j) = \alpha(i) + j\Delta x$, in percent; level 4 is terminal and carries none. Levels 3 and 4 have the same seven nodes. That is the tree refusing to widen once it reaches $\pm j_{\max}$.

At each level i the sum $\sum_j Q(i,j)\, e^{-r(i,j)\Delta t}$ returns $P(0,(i+1)\Delta t)$, the bottom row of the table, exactly, to the last bit.

Three features stand out. The Q's are not symmetric (0.0198 against 0.0205 at $i=2$) because the discounting differs across nodes even though the probabilities at j and $-j$ mirror each other. α rises along the curve, from 3.27% to 4.17%, because it is absorbing the upward slope of the forward rates. And the bottom node at level 3 carries −1.0232%: a Gaussian short rate goes negative, which Section 31.14 takes up as a property rather than a defect.

Nothing here is iterated. Equation (31.16) is one equation in one unknown with everything on the right already computed, which is why the curve comes back to machine precision rather than to a tolerance.

Run at $\Delta t = 0.25$ over eleven years, the same construction has 44 steps, $\Delta x = 0.008660$, $j_{\max} = 10$. Summing the Arrow–Debreu prices at each of those 44 levels returns the input discount factors with errors at or below one unit in the last place, and so does the 24-steps-a-year tree of the examples that follow. So the tree's discretization error touches option values, Section 31.9's subject, and never the curve: a mispriced swaption on this tree is never a

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symptom of a badly fitted curve.

31.9 BERMUDAN SWAPTIONS

Here is what the tree is for.

Example 31.8 One option with five dates, not five options

A receiver swaption on a swap paying 4% fixed annually and maturing at $t = 6$, notional 100, on the tree of Section 31.8 with $a = 0.08$ and $\sigma = 1\%$.

First check the machinery. A European receiver expiring in one year has a closed-form value by Jamshidian's decomposition (a European option on a coupon bond is a portfolio of options on the zeros), and that gives **1.1465**. The tree should converge to it:

Steps/year	Tree	Error
4	1.1637	+0.0172
8	1.1696	+0.0231
12	1.1667	+0.0202
24	1.1586	+0.0121
48	1.1502	+0.0037

The error shrinks as the grid is refined but not monotonically: eight steps a year is worse than four. Section 20.2 established why: a lattice's error depends on where the nodes fall relative to the payoff's kink, and refining the grid moves them. A desk that extrapolates two tree values without checking for this can get a worse answer than either: from four and eight steps, Richardson extrapolation gives 1.1755, further from 1.1465 than both.

Value the Bermudan exercisable annually at $t = 1, 2, 3, 4, 5$, at 24 steps per year, alongside each European:

Exercise right	Value
European, 1y only	1.1586
European, 2y only	1.1953
European, 3y only	1.0559
European, 4y only	0.7974
European, 5y only	0.4412
Sum of all five Europeans	4.6484
Bermudan, any one of the five	**1.6995**

The Bermudan is worth 42.2% more than the best single European and 37% of their sum. It exceeds any European because the right to choose has value. The holder does not have to guess in advance which date will be best.

It falls far short of the sum because the rights are substitutes, not additions. Exercising at year two extinguishes the option at years three, four, and five. A portfolio of five Europeans is five options; a Bermudan is one option with five dates, and the gap between 1.70 and 4.65 is the entire economic content of that sentence.

Black's model cannot produce the 1.6995. It prices each row of the top block, because each is a European option on a single rate. It has nothing to say about the choice, because making the choice requires knowing the curve at year three, and Black's model never builds one.

31.10 CALIBRATION, AND WHAT IT CANNOT DETERMINE

What calibration does here is narrower than the word suggests. Section 31.6 has already fitted the curve: $\theta(t)$ is solved so that today's $P(0, t)$ comes back exactly, and nothing about it is open to choice. What is left unfixed is the pair (a, σ), and those two govern volatility rather than level. So calibration in a fitted short-rate model means choosing two constants from option

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prices (the caps and swaptions of Section 29.9's volatility cube), never matching the curve.

The procedure is the ordinary one. Take a set of quoted instruments, price each on the model at a trial (a, σ), and minimize a weighted sum of squared differences,

$$\min_{a,\,\sigma} \sum_i w_i \left[V_i^{\text{model}}(a,\sigma) - V_i^{\text{market}} \right]^2$$

with the weights w_i chosen as Section 21.10 describes. The closed forms of Section 31.7 are what make this fast enough to run every morning, which is why they are derived.

Which instruments go into that sum decides what the calibration can see. Example 31.3 said a and σ are separately identified. The following example is the most important in this chapter, and it says that where it matters most, they are not.

Example 31.9 Every European matched, the Bermudan unmatched

Hold the one-year-into-five-year European receiver fixed at its tree value of 1.1586. For each mean-reversion rate a, solve for the σ that reproduces it exactly. Then reprice the Bermudan of Example 31.8.

a	σ	**European**	**Bermudan**	**vs.** $a = 0.08$
0.01	0.00823	1.1586	1.6474	−3.1%
0.03	0.00871	1.1586	1.6622	−2.2%
0.08	0.01000	1.1586	1.6995	—
0.15	0.01202	1.1586	1.7541	+3.2%
0.30	0.01724	1.1586	1.8705	+10.1%

Every row prices the calibration instrument identically, to four decimals, and the Bermudan spans 13.1% of its own value.

The mechanism is the reason two-factor models exist. A Bermudan's value depends on the *relationship* between rates at different exercise dates: how much the curve at year three tells you about the curve at year four. That relationship is governed by a, and a single European swaption contains no information about it, because it observes the curve once. There are three responses, and a desk should know which it is using.

Calibrate to a co-terminal strip. Use the swaptions that expire on each of the Bermudan's exercise dates and mature on its final date, which are exactly the Europeans the Bermudan chooses among. This is the standard practice, usually with σ piecewise constant so that every one is matched. It pins the volatility at each exercise date, but a, which governs how the dates relate, must still come from elsewhere: the second response.

Fix a from history and calibrate only σ. This is cruder, and defensible when the strip is not liquid. Section 29.9 reaches the same conclusion about SABR's β, which desks fix rather than fit: when the data cannot support a parameter, choose it outside the calibration rather than letting the fit invent it.

Or quote a range. Section 21.11 argued that model risk should be reported rather than hidden, and here the range is computable from a five-row calculation any desk can run.

What is not defensible is calibrating to one European and reporting one Bermudan price to four decimals, which the table shows to be four decimals of a number determined by an unobserved parameter.

31.11 CALLABLE BONDS AND OPTION-ADJUSTED SPREAD

The largest application of short-rate trees is not swaptions. It is the market for **callable bonds**, which the issuer may redeem at a fixed price, usually par, on a schedule of dates; the investor has sold the issuer a Bermudan receiver swaption in another form. And the number the model produces there has a name that overstates what it is.

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Example 31.10 What the Z-spread hides, and what the OAS assumes

A ten-year 5% annual bond, callable at par each year from year three, priced on the tree of Section 31.8 at 24 steps a year, as in Example 31.8.

Straight bond, zero spread	106.9013
Callable bond, zero spread	101.6743
Embedded call option	**5.2270**

Now suppose the bond trades at 99.00. Two spreads can be quoted. The **Z-spread** is the constant addition to the curve that reproduces the price while ignoring the call: **94.6 basis points**. The OAS (Section 29.2) is the constant addition that reproduces the price with the call modeled: **52.1 basis points**.

The 42.5 basis point difference is the call, expressed as a running yield. A portfolio manager comparing this bond with a non-callable one on Z-spread would see 94.6 and conclude it offered generous credit compensation. Roughly half of that is payment for an option the investor has written, and the OAS exists precisely to strip it out.

Now vary the model's parameters:

a	σ	**OAS**
0.03	1.0%	39.5 bp
0.08	1.0%	52.1 bp
0.20	1.0%	71.5 bp
0.08	1.5%	20.9 bp

The OAS ranges from 20.9 to 71.5 basis points across parameter choices that have not been calibrated to the same swaption prices. The Z-spread, for all its faults, is a fact about the price; *the OAS is a fact about the price and a model*, and the word "adjusted" does not convey how much of it is the second.

The practical rule follows immediately. OAS comparisons are meaningful *within* a book run on one model with one calibration, and close to meaningless across two houses that have not agreed on a and σ. Comparing a dealer's OAS with your own is comparing two models, not two bonds.

31.12 THE ONE-FACTOR DEFECT

Every model so far has one source of uncertainty, with a consequence that no amount of calibration can remove. At any future date, every rate on the curve is a deterministic increasing function of $r(t)$. A two-year rate and a ten-year rate cannot move in opposite directions, cannot move by unrelated amounts, and are perfectly correlated. The model does not merely approximate a correlation of one; it asserts it.

For a swaption or a callable bond this is tolerable, because the payoff depends on one rate or on a monotone function of the state. For anything whose payoff depends on the curve's *shape*, it is fatal, and it is straightforward to price how fatal.

Example 31.11 What perfect correlation costs

A five-year option on the spread between the ten-year and two-year swap rates is a **CMS spread option**, one of the most widely sold structured coupons. Forward rates 3.40% and 2.80%, normal volatilities 75 and 90 basis points a year, annuity 4.20 (an annual stream after expiration, simplifying the strip of caplets actually sold without changing the ratios). The forward spread is 60 basis points, and the option is struck at the money.

The spread's volatility is $\sqrt{\sigma_{10}^2 + \sigma_2^2 - 2\rho\,\sigma_{10}\sigma_2}$:

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ρ	Spread vol	Option value	Ratio
1.00	15.0 bp	56.2 bp	1.00
0.95	30.0 bp	112.4 bp	2.00
0.90	39.7 bp	148.7 bp	2.65
0.80	54.1 bp	202.6 bp	3.61
0.70	65.4 bp	245.0 bp	4.36
0.50	83.5 bp	312.9 bp	5.57

A one-factor model is forced to the first row. It returns 56.2 basis points—and the 15.0 is not even a correlation assumption; it is just $|\sigma_{10} - \sigma_2|$, the residual from two rates moving in lockstep at slightly different speeds. At a market-plausible correlation of 0.70 the option is worth 245.0 basis points, 4.4 times as much. Three things follow.

No calibration repairs this. One could force a one-factor model to reproduce the 245 by inflating σ, but then every swaption in the book would be mispriced. The error is in the model's structure, not its parameters, which is exactly the distinction Section 21.11 drew for equity models.

The error has a sign. A one-factor model *always* undervalues a spread option, because $\rho = 1$ minimizes the spread's volatility. A desk that sells these options at one-factor prices sells every one of them too cheaply.

The magnitude is the argument for Chapter 32. A factor of 4.4 is not a refinement.

31.13 TWO-FACTOR MODELS

The repair is a second source of uncertainty. The standard construction adds a stochastic mean-reversion level:

$$dr = \big[\theta(t) + u - ar\big]dt + \sigma_1\, dW_1, \qquad du = -b\,u\,dt + \sigma_2\, dW_2 \tag{31.17}$$

with $dW_1\, dW_2 = \rho\, dt$, where b is now the speed at which u reverts to zero rather than a level. Equivalently, r is the sum of two zero-mean mean-reverting factors and a deterministic shift chosen to fit the curve. This is the **G2++** form, whose name is a description of the construction: Gaussian, two factors, and "++" for the shift. The zero-mean form is how the model is usually implemented, because such factors discretize more cleanly.

Now the short rate and the long rate are driven by different combinations of two shocks, so rates at different maturities are imperfectly correlated and the curve can steepen, flatten, and twist. The model remains affine, so bond prices stay closed-form. The costs are real and should be weighed.

Five parameters instead of two, and the calibration is correspondingly less stable, a familiar problem from Section 21.4's account of Heston, where no parameter moved one feature alone.

And the lattice becomes two-dimensional. A two-factor tree is a 3×3 branching structure whose node count grows as the square, which is why two-factor implementations usually price by Monte Carlo or by finite differences on a two-dimensional grid rather than on a tree. There is a further obstruction on the tree: the simplest correlation correction to the nine branching probabilities is admissible only when the two G2++ factors, not the two shocks of (31.17), have a correlation no larger than $1/3$ in absolute value. [TN31] gives that obstruction and the change of variable that removes it, together with the two-factor bond and option formulas.

Section 32.17 sets out the rule for choosing among all these models; for the number of factors it comes down to whether the payoff depends on the level of rates or on their shape, and Example 31.11's question (what is the payoff worth at $\rho = 1$?) tells which.

31.14 NEGATIVE RATES AND THE REPAIRS

Hull–White's r is normally distributed, so it goes negative with positive probability, and Example 31.12 shows the probability is not small.

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Example 31.12 How often the model says rates are below zero

Take a flat forward curve at 3% and compute the probability the model assigns to a negative short rate, with $r(T)$ normal around the forward rate under the T-forward measure and with standard deviation $\sigma\sqrt{(1 - e^{-2aT})/(2a)}$:

	1y	2y	5y	10y
$a = 0.05$, $\sigma = 1.0\%$	0.1%	1.3%	6.5%	11.6%
$a = 0.05$, $\sigma = 1.5\%$	2.0%	6.9%	15.7%	21.3%
$a = 0.10$, $\sigma = 1.0\%$	0.1%	1.0%	4.6%	7.5%

At ten years the middle row assigns better than one chance in five to a negative short rate. Before rates went negative (in Denmark in July 2012, in the euro area in June 2014), this was cited as the model's principal flaw and as the reason to prefer CIR or a lognormal short rate.

Then rates went negative and stayed there for years, and the property became the reason Hull–White survived unchanged while its competitors, which could not be calibrated to that market without a shift, did not.

The general lesson is the one Section 19.9 drew about 1987 and Section 26.6 drew about house prices. A constraint imposed because the unconstrained case "cannot happen" is a statement about the sample, and the sample is always shorter than the future. A model's implausible region is where its next test comes from.

Where positivity or a floor is genuinely wanted, the standard repairs are a **shifted** model (apply a positive model such as CIR or a lognormal short rate to $r + s$ for a constant shift s, which moves the lower bound from zero to $-s$ rather than removing it) or the shifted and normal-versus-lognormal machinery of Section 29.8, which is the same idea applied to the quoting convention rather than to the process. A lognormal short rate such as Black–Karasinski, in which $\ln r$ mean-reverts, keeps positivity but has no closed-form bond price. Jamshidian's decomposition still holds, since the model is one-factor and monotone in its state, but it no longer yields a formula, and the tree's displacement $\alpha(i)$ must be found by Newton iteration at each step instead of from (31.16). That is the one genuine difference from Hull–White on the tree.

31.15 AFTER THE LIBOR TRANSITION

Chapter 7 rebuilt the curve framework around risk-free rates and Chapter 29 rewrote the caplet. What did that do to this chapter?

Less than one would expect, and for an instructive reason. A short-rate model models the instantaneous risk-free rate, which is closer to what SOFR and €STR actually are than to what LIBOR ever was. The transition removed an inconsistency rather than creating one: once discounting moved to OIS, and until the transition, the model's r was a risk-free rate while the products referenced a credit-sensitive rate, and the gap was absorbed by convention. Two things did change.

Discounting and forecasting are now the same curve for cleared trades in the overnight rates, so the multi-curve complication of Section 7.5 does not reach into the short-rate model at all. Where a spread is needed (a credit-sensitive benchmark, or a currency basis), it is added as a deterministic spread rather than modeled, which is usually adequate.

Compounded-in-arrears fixings also introduce a short averaging window that Section 29.4 handled by adjusting the caplet's variance. In a short-rate model it needs no separate adjustment: the model specifies the path of r, so the compounded rate over the accrual period is a functional of that path. In Hull–White $\int_{T_i}^{T_{i+1}} r\,ds$ is Gaussian, so the caplet on the compounded rate has a closed form, and in a simulation the integral is computed directly. On a recombining tree it

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needs one more state variable, since the tree does not remember the path.

31.16 WHERE THESE MODELS ARE STILL THE RIGHT TOOL

Chapter 32 introduces models that are more realistic in every respect and harder to use in most. The case for short-rate models is not nostalgia, and it rests on four properties.

They price American and Bermudan exercise naturally. A tree gives the continuation value at every node, so an early-exercise decision costs nothing extra. A forward-rate model does not, and needs the regression machinery of Section 20.11. For the callable and Bermudan books, which are among the largest users of interest-rate models by notional, this is decisive.

They are fast. A short-rate tree prices a Bermudan in milliseconds, which matters when a desk revalues thousands of positions and computes their sensitivities every night.

They have few parameters, which is a defect when the market is rich and a virtue when it is not. Example 31.9 showed that two parameters were already one too many for the data.

And they model path dependence in the short rate directly, which Section 31.15 illustrated with compounded fixings.

31.17 WHO USES THEM, AND FOR WHAT

Callable-bond desks and the investors who buy their paper. The OAS of Example 31.10 is produced by a short-rate model at most institutions that quote one, and Section 26.3's mortgage prepayment models sit on top of the same machinery: a prepayment model needs a rate path to respond to, and this is what generates it. In agency mortgage-backed securities OAS is the standard relative-value measure.

Bermudan swaption desks. The supply comes from callable debt: an agency, supranational or corporate issuer that swaps a callable bond to floating with a cancellable swap leaves the dealer long a Bermudan receiver option, which it sells on to mortgage investors, servicers and banks hedging prepayment, all short the same option through their borrowers. Heavy callable issuance therefore weighs on long-dated swaption volatility, and for the desk in between Example 31.9's calibration question is a daily decision.

Insurers and pension funds valuing liabilities with embedded guarantees: a minimum crediting rate is an interest rate floor with a fifty-year tenor, and it must be valued on a model that can generate a coherent curve that far out.

Structured-note issuers, for the level-dependent coupons but *not for the spread-dependent ones*, which is the division Section 31.13 drew and the reason a single desk often runs both a one-factor and a two-factor model and routes trades between them.

And risk managers, for scenario generation, where, unusually, the equilibrium models of Section 31.2 come back, because the question being asked is what could happen rather than what a hedge costs.

31.18 WHERE THIS CAME FROM

The apparatus Chapter 15 built for equity options does not transfer to bonds, because a bond's volatility must vanish at maturity and because the whole curve, not one price, is the state variable. Sorting that out took twenty years.

Vasicek (1977) is the first serious answer: a mean-reverting short rate from which the entire curve follows in closed form. Its defect is fundamental and instructive—being an equilibrium model it produces a curve, and that curve is not today's. Cox, Ingersoll and Ross (1985) gave the approach its general-equilibrium foundation and a square-root diffusion that keeps rates positive, and the same defect remained.

Ho and Lee (1986) inverted the problem: take today's curve as given and fit the model to it. That is the no-arbitrage turn, and everything since sits on the far side of it. Hull and White

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(1990) combined the two, adding mean reversion to a fitted model, which is the formulation this chapter uses; Black, Derman and Toy (1990) and Black and Karasinski (1991) did the same in lognormal form. Jamshidian (1989) showed that an option on a bond portfolio decomposes into a portfolio of options in a one-factor model, which is why European swaptions had closed forms in these models from the start; Bermudans waited for the lattice.

SUMMARY

Black's model prices one option on one rate, and cannot price a Bermudan swaption or a callable bond, which need the whole curve at each future decision date. A short-rate model generates that curve from a single state variable, the source of both its tractability and its central defect. A handful of parameters cannot match a curve with hundreds of points, so the no-arbitrage generation makes the drift a solved function of time: $\theta(t)$ carries no information, and every model in this class is an interpolation device with a stochastic component. CIR's $\sqrt{r}$ keeps rates positive, which is why it could not be calibrated once rates went negative.

In Hull–White, σ sets the level of the volatility term structure and a sets its shape, forward-rate volatility decaying as σe^{-aT}. The tree reprices the curve to machine precision, because the displacement $\alpha(i)$ is solved rather than fitted, and its option values converge as the grid is refined, but not monotonically.

A Bermudan exercisable at several dates is worth clearly more than the best single European and far less than the sum of all of them: a portfolio of Europeans is several options, a Bermudan is one option with several dates. Holding the calibrating European fixed and varying a leaves every European identical and still moves the Bermudan materially, because a single European observes the curve once and contains no information about a. Calibrating to one European and quoting the Bermudan to four decimals is not defensible.

On a callable bond the Z-spread exceeds the OAS by enough that much of the apparent credit compensation pays for a written option, but the OAS itself moves widely across parameter choices that no common calibration has fixed. The Z-spread is a fact about the price; the OAS is a fact about the price and a model, so comparing OAS across houses compares models rather than bonds.

In one factor every rate is an increasing function of one state variable, so all rates are perfectly correlated—asserted, not approximated. A CMS spread option is forced to $\rho = 1$ and is worth a fraction of its value at a realistic correlation; no calibration repairs it, and the error always has the same sign. Use one factor for level products and two for shape products, and test a payoff by what it is worth at $\rho = 1$.

Gaussian rates go negative, with a probability far from negligible at long horizons. Cited as the model's principal flaw before rates went negative, this property became the reason it survived: a model's implausible region is where its next test comes from. Short-rate models remain right for early exercise, for speed, and when the data cannot support more parameters—and wrong whenever the payoff depends on the curve's shape.

FURTHER READING

Vasicek's and Cox, Ingersoll, and Ross's original papers are short and state what an equilibrium model is trying to do; reading them after this chapter is more useful than before it, because the thing they are trying to do is the thing Section 31.4 says a desk cannot use. Hull and White's papers on the extended model and on the trinomial construction are the primary sources for Sections 31.6 and 31.8, and the tree paper describes the two-stage procedure precisely enough to code from. Brigo and Mercurio's book is the standard reference for the whole family, including the G2++ form of Section 31.13 and the shifted variants of Section 31.14, and it is where to go for the algebra this chapter states without deriving. Andersen and Piterbarg's three

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volumes are the practitioner's reference and treat the calibration question of Example 31.9 at much greater length, reaching the same conclusion. On OAS, Fabozzi's *Handbook of Fixed Income Securities* sets out the parameter dependence of Example 31.10 from the holder's side (what a quoted spread assumes about volatility and the exercise policy), while Andersen and Piterbarg's chapters on callable products document the models the quotes come from; read the first to interrogate a quoted OAS, the second to reproduce one.

PRACTICE QUESTIONS

31.1 Explain why Black's model cannot price a Bermudan swaption, and be specific about what it fails to produce.

31.2 In Vasicek's model with $a = 0.10$, $b = 5\%$, $\sigma = 1\%$, compute the limiting long rate and explain why it is below b. Then explain why an equilibrium model cannot match an observed term structure, and why this is arithmetic rather than calibration.

31.3 Explain what $\theta(t)$ does in Ho–Lee, why it carries no information, and state the defect that rules the model out in practice together with the quantity that exposes it.

31.4 Using σe^{-aT}, compute the ten-year forward-rate volatility at $\sigma = 1\%$ for $a = 0.05$ and $a = 0.20$, and explain which feature of the volatility term structure is set by a and which by σ.

31.5 Describe the two stages of the Hull–White tree construction, state which quantities are solved rather than fitted, and explain why the tree reproduces the initial curve to machine precision while still mispricing options.

31.6 Explain why the branching pattern must change at $\pm j_{\max}$, and what would go wrong if it did not.

31.7 In Example 31.8, explain why the Bermudan is worth more than any single European and much less than their sum, and why the tree's convergence to the Jamshidian value is not monotone.

31.8 Explain why a single European swaption contains no information about the mean-reversion rate, use Example 31.9 to state the range of Bermudan values consistent with an exactly matched European, and describe three defensible responses and one that is not.

31.9 With $a = 0.08$, $\sigma = 1\%$ and $\Delta t = 1$, compute Δx, $j_{\max}$, the branching probabilities at $j = 2$ and $j = 3$, and $\alpha(0)$ from $P(0, 1) = 0.967873$, and check them against Example 31.7.

31.10 Explain the difference between a Z-spread and an option-adjusted spread and what the gap between them measures; then, using Example 31.10, state the OAS at $a = 0.08$ with $\sigma = 1.0\%$ and with $\sigma = 1.5\%$, and explain what this implies about comparing OAS across dealers.

31.11 Explain why a one-factor model forces perfect correlation between all rates, and why this is an assertion rather than an approximation.

31.12 Using Example 31.11, state the CMS spread option's value at $\rho = 1$ and at $\rho = 0.70$, and explain why the one-factor error always has the same sign and why that sign is the dangerous one.

31.13 State the decision rule for choosing between a one-factor and a two-factor model, and give one product on each side of it.

31.14 Explain why Gaussian short rates were regarded as defective before rates went negative and as necessary afterward, and name four properties that keep short-rate models in use together with the single property that rules them out.

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31.15 Using Example 31.5, state the caplet's value under the forward measure and under the risk-neutral measure and comment on their agreement, then explain what the 2.52% error in part (c) consists of and its sign.

31.16 Using Example 31.6, state the range of $E^{Q}[S_T]/F(0,T)$ across the correlations shown, explain what quantity produces it, and name the two other results in this book that are the same quantity.

FURTHER QUESTIONS

31.17 Example 31.1 shows three curves converging to $R_\infty = b - \sigma^2/(2a^2)$. Derive this limit, explain the convexity argument that produces the second term, and determine the parameter values at which the correction becomes large enough to matter for a thirty-year bond.

31.18 Example 31.9 shows the Bermudan spanning 13.1% of its value with the European held fixed. Extend the calculation to the co-terminal strip: for each a from 0.01 to 0.30, bootstrap a piecewise-constant $\sigma(t)$ that matches all five co-terminal Europeans exactly, reprice the Bermudan, and explain why the range does not fall to zero and what would narrow it.

31.19 Example 31.10 shows the OAS moving from 20.9 to 71.5 basis points. Propose a reporting convention that would make OAS comparable across institutions, state what each party would have to disclose, and explain why the market has not adopted one.

31.20 Example 31.11 prices the one-factor defect at a factor of 4.4. Derive the spread's volatility as a function of ρ, show that $\rho = 1$ minimizes it for any $\sigma_{10} \neq \sigma_2$, and determine the conditions under which a one-factor model could be made to price a single spread option correctly, and explain what it would then get wrong.

31.21 Section 31.13 says a two-factor model costs five parameters and a two-dimensional lattice. Compare this trade-off with Section 21.4's account of Heston, where additional parameters also failed to move features independently. State the general principle governing when an extra factor is worth its calibration cost, and apply it to both cases.

31.22 Section 31.14 argues that a model's implausible region is where its next test comes from, citing 1987 and house prices. Identify a constraint currently imposed on rate models because the alternative "cannot happen," argue both sides, and propose how a desk should hedge the possibility that it can.

31.23 Section 31.16 claims short-rate models remain right for Bermudan exercise while Chapter 32's models are more realistic. Construct the strongest case that a forward-rate model with American Monte Carlo should replace them for the callable book, state what would have to be true about accuracy and computation for that case to win, and give your own judgment with reasons.

31.24 Example 31.5 finds a 2.52% error from ignoring the payoff-discount covariance. Derive that covariance term analytically for a caplet under Hull–White, determine its dependence on rate volatility and option maturity, and identify the configuration in which it would exceed 10%.

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